

7.2 Mixture of Experts

A router sends each token to a few experts. Capacity without dense FLOPs.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

A dense transformer uses **every** MLP on every token. A **mixture of experts (MoE)** keeps many MLPs (**experts**) and runs only a few of them per token. Capacity grows with the expert count; FLOPs per token stay closer to a small dense net.

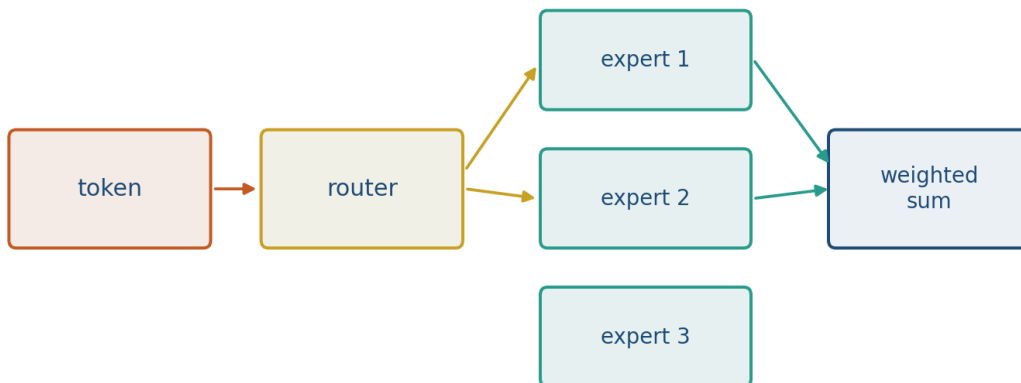
1. Router plus experts

Replace the block MLP with E experts $E_1, \dots, E_E$ and a **router** (a linear map plus softmax or a noisy top- k). For token x ,

$$y = \sum_{i \in \mathcal{T}(x)} g_i(x) E_i(x),$$

where $\mathcal{T}$ is the set of k selected experts (often $k = 1$ or 2) and g_i are the gate scores. Attention can stay dense; the sparse part is the feedforward path, where most parameters already live (note **3.3**).

Sparse MoE: most experts stay off for a given token



Switch Transformer, GLaM, Mixtral: same picture, different k , expert size, and load-balancing tricks.

2. Why sparsity, and what breaks

Sparse means most experts are idle for a given token. That is the point: you can store $8 \times$ the MLP weights and still pay about 2 experts of compute. Training needs a **load-balancing** term so the router does not send every token to expert 0. At inference, experts must sit in RAM or be paged; a “small” Mixtral still has a large footprint.

Routing is discrete. Gradients flow through the chosen experts and through the gate (straight-through or a softmax over a shortlist). Expert collapse and token dropping are the usual failure modes.

3. Dense versus MoE in a project

Use a dense 7B if you need simple deployment. Use an MoE if you need more capacity at a similar token-FLOP budget and you can host the weights. Do not fine-tune one expert and claim the mixture improved; say whether you trained the router. Week 7.3 is how you shrink whatever you picked.

4. Practice

1. If $k = 1$ and the router always picks expert 2, what did you just train?
2. Why can an MoE with 8 experts have roughly the compute of 2 dense MLPs per token, not 8?
3. Name one reason Mixtral-style serving is harder than a dense model with the same active FLOPs.